Qualification of Pathology Foundation-Model Signals Across Cohorts, Sites and Clinical Endpoints

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Abstract

Pathology foundation models make image–label associations easy to detect, but high pooled performance does not establish whether signals transport or add information beyond clinical covariates. We tested the proposition that signals require evidence for cohort transport, confounding, site behavior, representation stability, and endpoint fidelity. We operationalized these requirements as a qualification framework with three states—transportable, context-sensitive, and unsupported in the frozen design—and applied it to frozen CONCH and Virchow signals for grade, phenotype, PTEN, SPOP, androgen-receptor activity, and post-hoc recurrence across five prostate-pathology resources. Grade and phenotype had the strongest cross-cohort support. PTEN remained detectable within TCGA-PRAD but lacked a supported held-out increment beyond grade. Androgen-receptor activity retained a positive pooled direction while site transport remained unresolved. SPOP was unsupported in the frozen primary configuration and setting-sensitive. Recurrence estimates varied by endpoint, encoder, and scale and lacked established incremental value beyond fuller clinical and site covariates. Thus, pooled performance separated into evidence states rather than a model ranking. This study contributes a reusable qualification framework and empirical failure map that expose when apparently successful probes are transportable, conditional, or unsupported, helping prioritize signals for further validation.

1 Introduction

Pathology foundation models turn whole-slide images into reusable representations from which lightweight probes can recover grade, tissue phenotype, molecular labels and clinical outcomes [1, 2]. This capability has made candidate signals much easier to produce than to interpret. A high pooled AUROC or correlation can arise because the representation captures the target, but it can also reflect grade composition, institutional case mix, acquisition conditions, sampling choices or the definition of the clinical endpoint. Consequently, the central problem is no longer only whether a frozen embedding contains a statistically detectable association. It is whether the available evidence is sufficient to qualify that association as transportable, to restrict it to a specific context, or to leave it unsupported in the evaluated design.

This distinction matters because common validation practices collapse different scientific questions into a single performance number. External transport asks whether a signal retains direction in a different cohort without refitting. Confounder auditing asks whether it contributes information beyond established clinical structure such as grade. Site analysis asks whether a pooled result survives institutional heterogeneity. Representation and sampling audits ask whether the interpretation depends on encoder, physical scale, tile budget or sampling seed. Endpoint auditing asks whether an outcome association survives a change in event definition rather than merely

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a change in name. Success on one axis cannot substitute for missing evidence on another, and repeated estimates obtained from the same patients and folds are sensitivity analyses rather than independent replications.

We organized the study around three testable propositions. First, signals corresponding to morphology directly expressed in routine histology should show stronger cross-cohort transport than signals whose labels are molecular or outcome-derived. Second, an apparent molecular or outcome association may narrow after grade, site, representation or endpoint audits even when its pooled estimate is positive. Third, candidate signals should therefore be assigned evidence states from the pattern of support across these dimensions, not ranked or validated by one aggregate metric. These propositions define the logic evaluated here; they do not imply that every target had an identical external cohort or that lack of support in the frozen analysis establishes biological absence.

To test this logic, we assembled complementary evidence from five prostate-pathology data resources. NADT-Prostate provided source-cohort grade and phenotype evaluations; PANDA provided case images from Karolinska and Radboud for transfer without refitting [3]; PRECISE provided an additional session-level grade evaluation; TCGA-PRAD provided molecular labels, site structure and explicitly distinguished recurrence endpoints; and the LEOPARD-derived analysis supplied a post-hoc recurrence signal for transfer. Across frozen CONCH and Virchow representations, we evaluated grade, tumor phenotype, PTEN loss, SPOP mutation, androgen-receptor activity and recurrence. The analyses linked cross-cohort transfer to held-out confounder increments, site-specific uncertainty, a correlated encoder–scale–tile–seed grid and endpoint-specific survival comparisons. This design allowed the same candidate to succeed on one evidentiary axis while remaining unresolved on another.

The study makes three contributions. First, it operationalizes pathology foundation-model signal qualification as a joint, target-level assessment of transport, confounding, site behavior, representation stability and endpoint fidelity. The resulting categories— transportable, context-sensitive and unsupported in the frozen design—describe the scope of evidence rather than a league table of model performance. Second, it provides an empirical evidence map across morphologic, molecular and outcome targets: grade and phenotype showed the clearest transport; PTEN and androgen-receptor activity required narrower interpretations; SPOP was unsupported in the fixed primary configuration; and recurrence remained endpoint- and representation-conditioned. Third, it demonstrates why pooled discrimination alone is insufficient: apparently positive probes can lose incremental support, remain unresolved across sites or change interpretation with endpoint and representation. The accompanying source-linked workflow makes those qualification decisions auditable and reproducible. The contribution is therefore not another encoder or a deployable classifier, but a concrete framework and empirical failure map for deciding which foundation-model signals warrant further validation, under what conditions, and with which claims.

2 Results

2.1 Qualification framework converts the propositions into evidence tests

Analysis frame. Candidate signals were evaluated at their fixed analysis unit whenever an endpoint was available. The Results follow the evidentiary sequence introduced above. We first test whether morphology-linked signals transport across cohorts, then ask whether molecular and outcome signals survive confounder, site, representation and endpoint audits, and finally integrate those findings into target-specific evidence states. Each claim is linked to its cohort, endpoint, frozen encoder output and saved source table (Figure 1).

Evidence test. Five dimensions were kept distinct: cross-cohort transport, grade-adjusted increment, site behavior, correlated setting sensitivity and endpoint dependence. They produced three submission-facing states: transportable, context-sensitive and unsupported in the frozen design. Support on one dimension did not substitute for missing evidence on another. The complete target-by-axis coverage audit is provided in the supplementary evidence matrix and its source-linked audit table.

Interpretive boundary. Not every target had a second cohort with the same label, and setting comparisons reused patients and folds. The framework therefore qualifies the evidence available for each target; it does not assume equal validation opportunities or convert statistical non-support into biological absence.

Qualification rule. No pooled metric alone qualified a signal. A target's state was assigned only after the relevant evidentiary axes had been considered together.

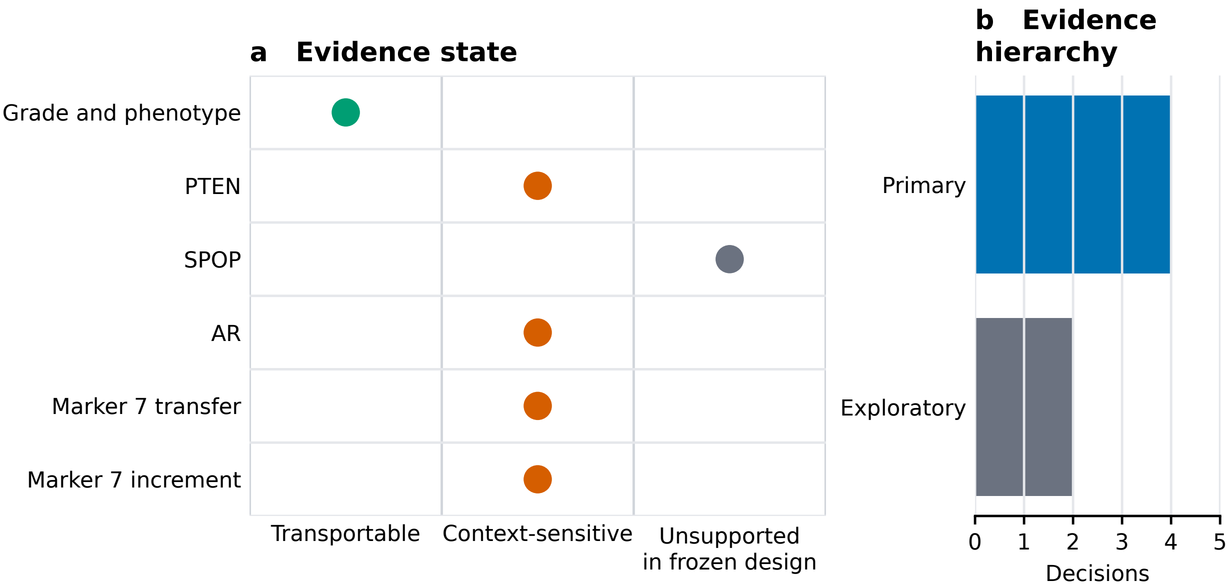

Figure 1: Qualification framework and evidence map. **a**, Each dot assigns one target-level decision to the strongest evidence state supported by the frozen analyses: transportable, context-sensitive, or unsupported in the frozen design. **b**, Counts of the same six decisions by evidentiary hierarchy comprise four primary decisions and two exploratory marker-7 decisions. The states summarize the joint pattern across cross-cohort transport, confounder adjustment, site behavior, correlated setting sensitivity, and endpoint dependence; they are not a model-performance ranking, and support on one axis does not replace evidence missing on another.

2.2 Morphology-linked signals show the strongest cross-cohort transport

Analysis frame. The frozen NADT-Prostate CONCH probes were evaluated on patients in the source cohort and then applied without refitting to PANDA case images from Karolinska and Radboud. PANDA used a 100-case cap per institution–ISUP stratum: 1,200 sampled cases were drawn and 1,137 passed tissue filtering. The phenotype label defined ISUP 0 as benign and ISUP ≥ 1 as tumor. A separate PRECISE analysis used evaluable imaging sessions with pathologist-assigned Gleason scores (Figure 2).

Evidence state. Among 39 NADT-Prostate patients, the patient-level Gleason correlation was 0.478 (95% percentile patient-bootstrap interval 0.170–0.712; 2,000 requested resamples; valid and undefined replicate counts were not retained in the saved legacy output), and the phenotype correlation was 0.800 (95% percentile patient-bootstrap interval 0.625–0.909; 2,000 requested resamples; valid and undefined replicate counts were not retained in the saved legacy output). In PANDA, Gleason correlations were 0.354 for 565 Karolinska cases and 0.519 for 572 Radboud cases; phenotype AUROCs were 0.786 and 0.871, respectively, with 469/565 tumor cases at Karolinska and 483/572 tumor cases at Radboud. In PRECISE, the session-level Gleason correlation was 0.865 across 17 evaluable sessions. These consistent cross-cohort directions supported a transportable state for grade and phenotype.

Local limitation. The saved source included no intervals for the PANDA and PRECISE rows. PANDA's capped sampling and grade-derived phenotype label constrain interpretation of its case-image estimates, and repeated encoder, scale and tile settings reused the same underlying cohorts.

Qualification decision. Grade and phenotype provided the strongest evidence of cross-cohort transport among the evaluated targets. This supports the first proposition within the available cohorts, while remaining specific to these morphology-linked targets and analysis units.

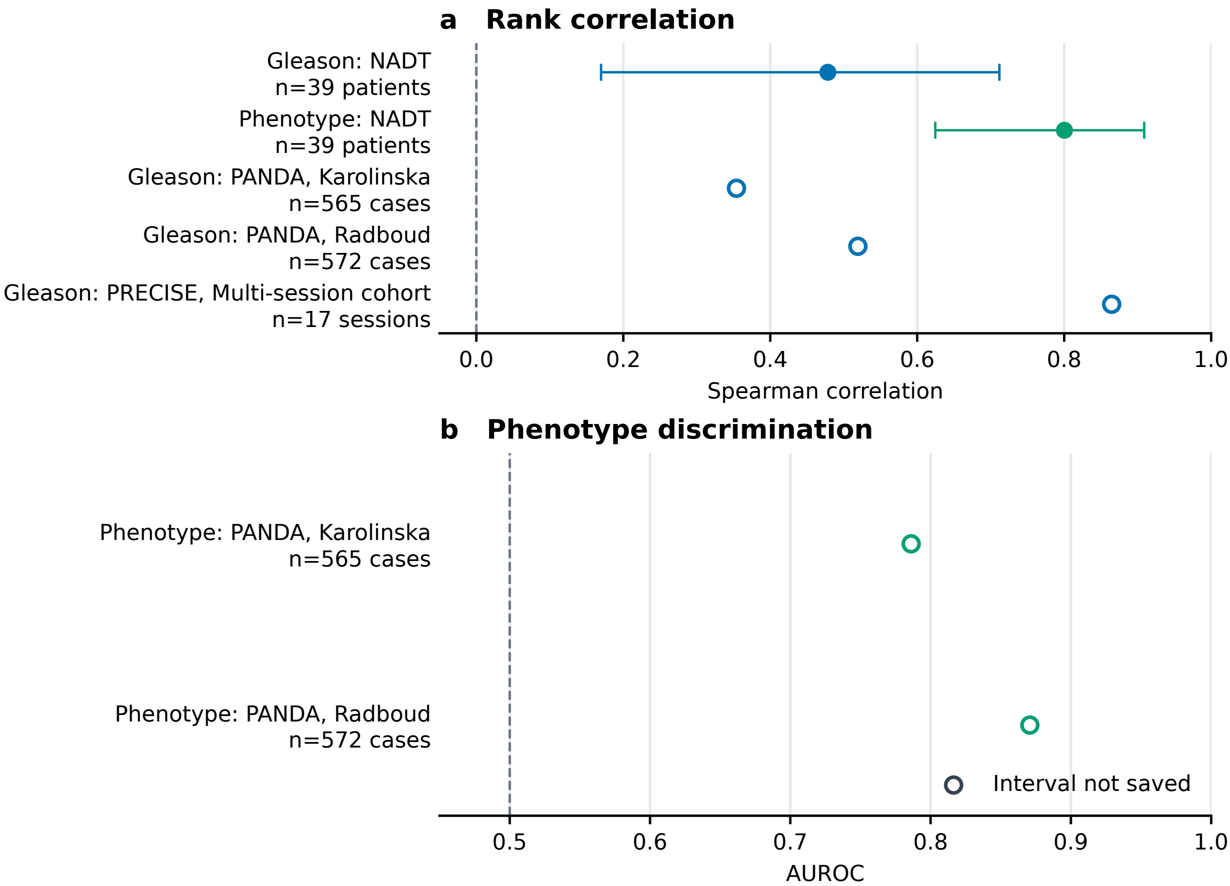

Figure 2: Cross-cohort grade and phenotype results. **a**, Spearman rank correlations for Gleason or phenotype in NADT-Prostate and for Gleason after transfer without refitting to PANDA and PRECISE. **b**, Phenotype AUROC after transfer without refitting to the Karolinska and Radboud PANDA subsets. Filled circles with horizontal whiskers show the saved point estimate and patient-bootstrap interval; open circles mark estimates for which no interval was saved. Dashed vertical lines denote the metric null (0 for Spearman correlation and 0.5 for AUROC). Labels give the saved patient, case-image, or session denominator; unavailable intervals were not reconstructed.

2.3 Pooled molecular associations do not determine qualification

Analysis frame. PTEN loss, SPOP mutation and androgen-receptor (AR) activity were evaluated at the patient level in the fixed 273-patient TCGA-PRAD frame using frozen CONCH outputs; the correlated configuration summaries additionally included Virchow (Figure 3).

Evidence state. PTEN had a frozen-primary AUROC of 0.632 (patient-bootstrap interval 0.560–0.706), and all evaluated cells were above chance, with a global seed-cell range of 0.519–0.717. AR had a frozen-primary Spearman correlation of 0.195 (0.074–0.310) and a global seed-cell range of 0.136–0.297. SPOP had a frozen-primary AUROC of 0.519 (0.408–0.627), while its global seed-cell range crossed chance (0.348–0.679). PTEN and AR were therefore context-sensitive; SPOP was unsupported in the frozen primary design and configuration-sensitive in the broader audit. The site-stratified SPOP audit did not overturn that decision (Figure 3c): CH contained no SPOP-positive patients and therefore had an undefined AUROC, while the saved patient-bootstrap interval at each of the other five source-coded sites included the AUROC null.

Local limitation. PTEN and SPOP event counts were not recorded in the saved primary rows, and TCGA-PRAD did not provide cross-cohort qualification for these molecular labels. The SPOP range neither excludes a modest effect nor establishes biological absence.

Qualification decision. The pooled molecular results generated different questions rather than final claims: PTEN required an increment audit, AR required site qualification, and SPOP remained unsupported under the fixed primary design. This is the first test of the second proposition that a positive or apparently stable pooled estimate can narrow under target-appropriate audits.

2.4 Confounder and site audits narrow the PTEN and AR claims

Analysis frame. Grade-adjusted increments were estimated on held-out TCGA-PRAD patients for CONCH and Virchow. AR site analyses used slide-level correlations with patient-cluster bootstrap intervals, whereas the pooled AR row used a patient-level estimate (Figure 4).

Evidence state. For PTEN, CONCH's held-out increment beyond grade was +0.019 (patient-bootstrap interval −0.025 to +0.068); the Virchow increment was +0.035 (−0.013 to +0.087). The corresponding AR increments in R^2 were +0.004 (−0.036 to +0.042) and +0.019 (−0.015 to +0.051). The pooled AR correlation was positive, but every stored site-specific interval crossed zero.

Local limitation. The site rows combine a slide-level effect estimate with patient-cluster resampling, and site remains entangled with acquisition metadata. Binary endpoint event counts were not available in the saved PTEN increment rows. No site or scanner cause can be assigned from these comparisons.

Qualification decision. Grade-independent PTEN and AR increments were not established. AR's pooled direction was supported, but site transport remained unresolved. The confounder and site audits therefore narrowed the interpretation of the pooled signals, directly supporting the second proposition.

2.5 Recurrence transfer changes with endpoint and covariate hierarchy

Analysis frame. The post-hoc marker 7 risk score was transferred from LEOPARD to a fixed 270-patient TCGA-PRAD target frame. Frozen-risk performance was evaluated separately for the Reconstructed with-tumor endpoint and Official TCGA-CDR PFI. Increment analyses used

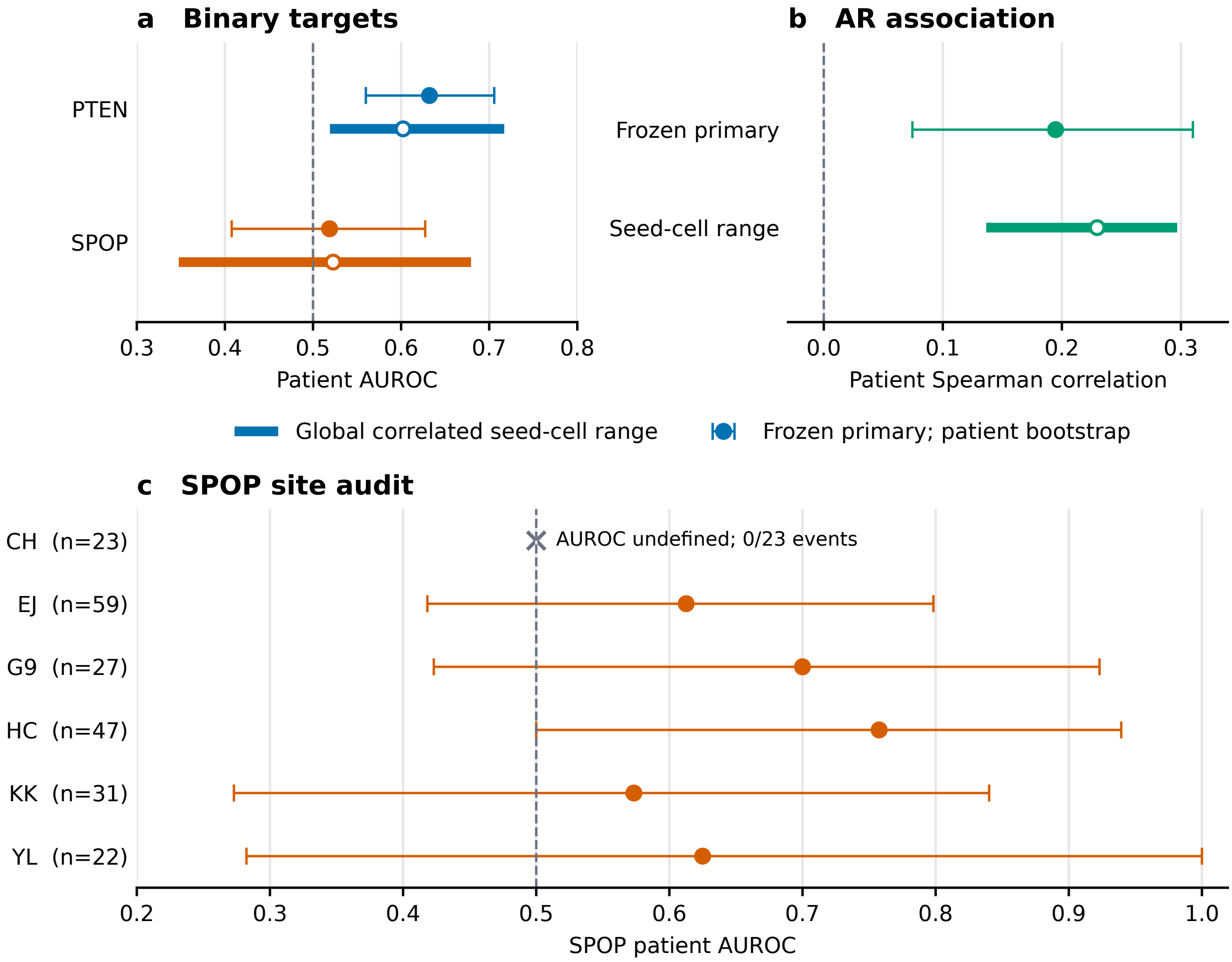

Figure 3: Patient-level molecular qualification. **a**, PTEN-loss and SPOP-mutation patient AUROCs in the fixed 273-patient TCGA-PRAD frame. **b**, Androgen-receptor (AR) activity patient-level Spearman correlation in the same frame. In **a**–**b**, filled circles and whiskers are frozen-primary estimates and saved patient-bootstrap intervals; thick lines are the global minimum–maximum range across 60 correlated encoder–scale–tile–seed cells, with open circles marking their mean. These ranges are sensitivity summaries, not patient-level confidence intervals. **c**, site-stratified SPOP patient AUROCs and patient-bootstrap intervals; site codes are retained from the source. The cross marks CH, where AUROC was undefined because there were no positive events among 23 patients. Dashed lines mark 0.5 for AUROC and 0 for correlation.

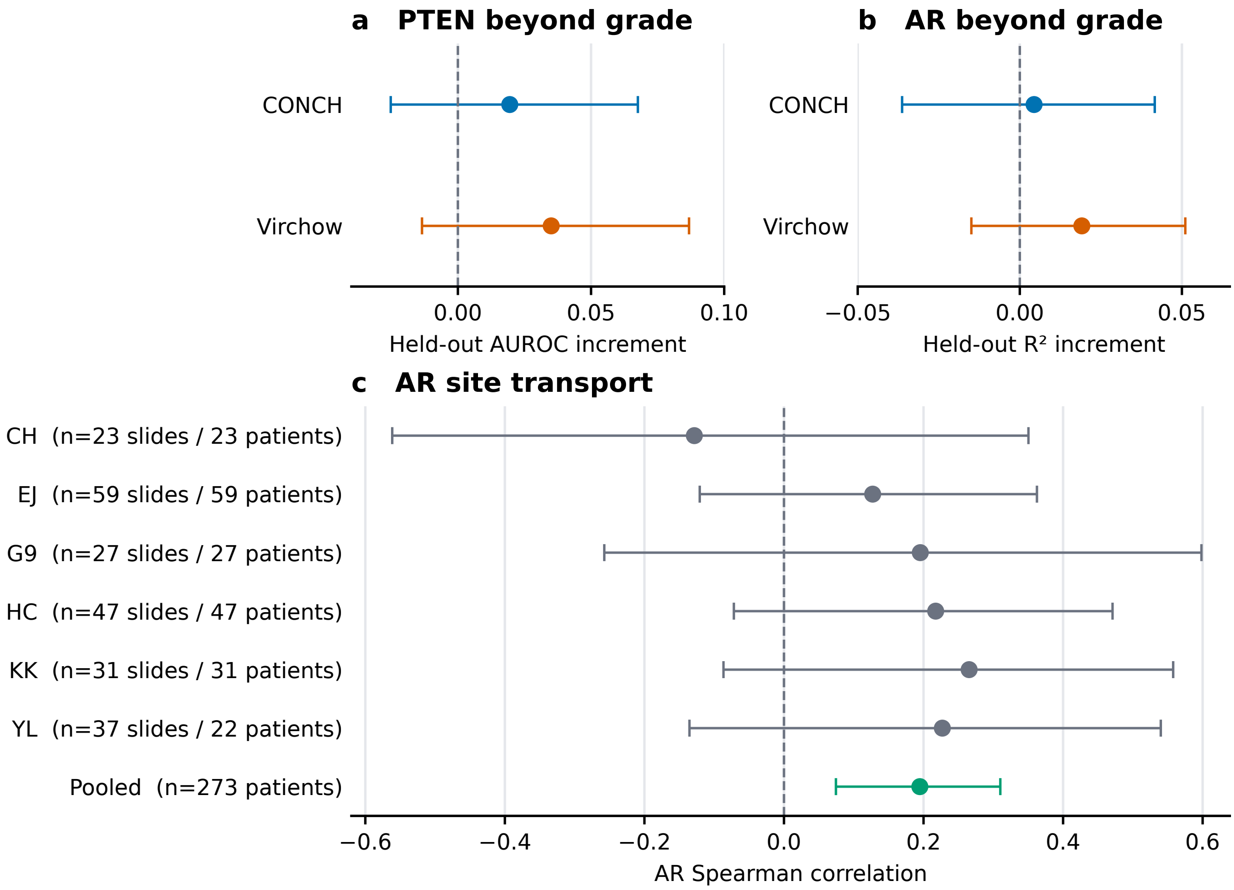

Figure 4: Confounder and site audits. **a**, Held-out patient-level PTEN AUROC increment beyond grade for CONCH and Virchow. **b**, Corresponding held-out AR R^2 increment. Both increment panels use the fixed 273-patient frame; points and whiskers are saved patient-bootstrap estimates and intervals. **c**, Site-specific slide-level AR Spearman correlations with patient-cluster bootstrap intervals, followed by the pooled patient-level estimate with its patient-bootstrap interval. Source site codes and slide/patient denominators are printed on the ordinate. Dashed vertical lines mark zero; positive increments favor adding the image signal. These observational site comparisons do not identify scanner, stain, or site as a cause.

the same 153-patient complete-case cohort, with 30 reconstructed-endpoint events and 15 official PFI events (Figure 5).

Evidence state. For the reconstructed endpoint, frozen-risk C-index was 0.673 (95% CI 0.587–0.759; 270 patients, 57 events). The official PFI C-index was 0.586 (95% CI 0.482–0.689; 270 patients, 42 events). The reconstructed endpoint grade-plus-image versus grade delta C-index was +0.083 (95% CI +0.012 to +0.157), but its paired integrated-Brier-score interval included zero. The official PFI grade-plus-image increment was +0.032 (−0.040 to +0.126), and all official-PFI paired C-index and IBS intervals include zero.

Local limitation. Marker 7 was discovered post hoc, endpoint definitions differed, and multiple imputation was not performed. The official-PFI result and every paired comparison were based on the saved evaluable or complete-case frames; unobserved outcomes were not treated as events or non-events.

Qualification decision. The recurrence signal remained exploratory and context-sensitive. Its effect and increment depended on endpoint and covariate hierarchy, and the full clinical/site comparisons did not support a general incremental claim. This provides an outcome-level demonstration of the second proposition.

2.6 Correlated setting audits distinguish stable from conditional signals

Analysis frame. Six markers were each summarized across 12 encoder–scale–tile configurations and five sampling seeds, giving $6 \times 12 \times 5 = 360$ correlated seed cells. Marker 7's setting sensitivity was interpreted alongside its endpoint-conditioned transfer summary (Figure 5), and the decision-level grid summary is shown in Figure 6. Paired contrasts preserved shared patients, folds and settings.

Evidence state. No configuration's observed five-seed range straddled the metric null for Gleason, phenotype, PTEN or AR. The observed five-seed range straddled the null for SPOP in 8/12 configurations and marker 7 in 1/12. Across the 60 seed cells for each marker, the global seed-cell range was 0.271–0.646 for Gleason and 0.845–0.965 for phenotype; PTEN remained above its null across all seed cells. SPOP and marker 7 showed greater setting sensitivity than the transported signals.

Local limitation. The 360 cells are correlated sensitivity settings, not separate patient samples or repeated confirmatory tests. A configuration's observed seed range and its descriptive Student-t interval across seeds are different summaries; the 8/12 and 1/12 counts refer only to observed-range straddling. Neither summary quantifies population uncertainty, and paired contrast directions do not identify a causal effect of encoder, scale or tile budget.

Qualification decision. The stability audit bounded setting dependence without elevating evidence states. It reinforced transported directions for grade and phenotype, preserved within-cohort PTEN support, and narrowed SPOP and marker 7 to setting-conditioned interpretations. Repeated performance across correlated settings therefore informed robustness but did not constitute independent validation.

2.7 Joint evidence states replace a single performance ranking

Across targets, the three propositions converged on one result: detectability and qualification were not interchangeable. Morphology-linked grade and phenotype provided the clearest transport evidence. Molecular signals separated after confounder and site audits, and the recurrence

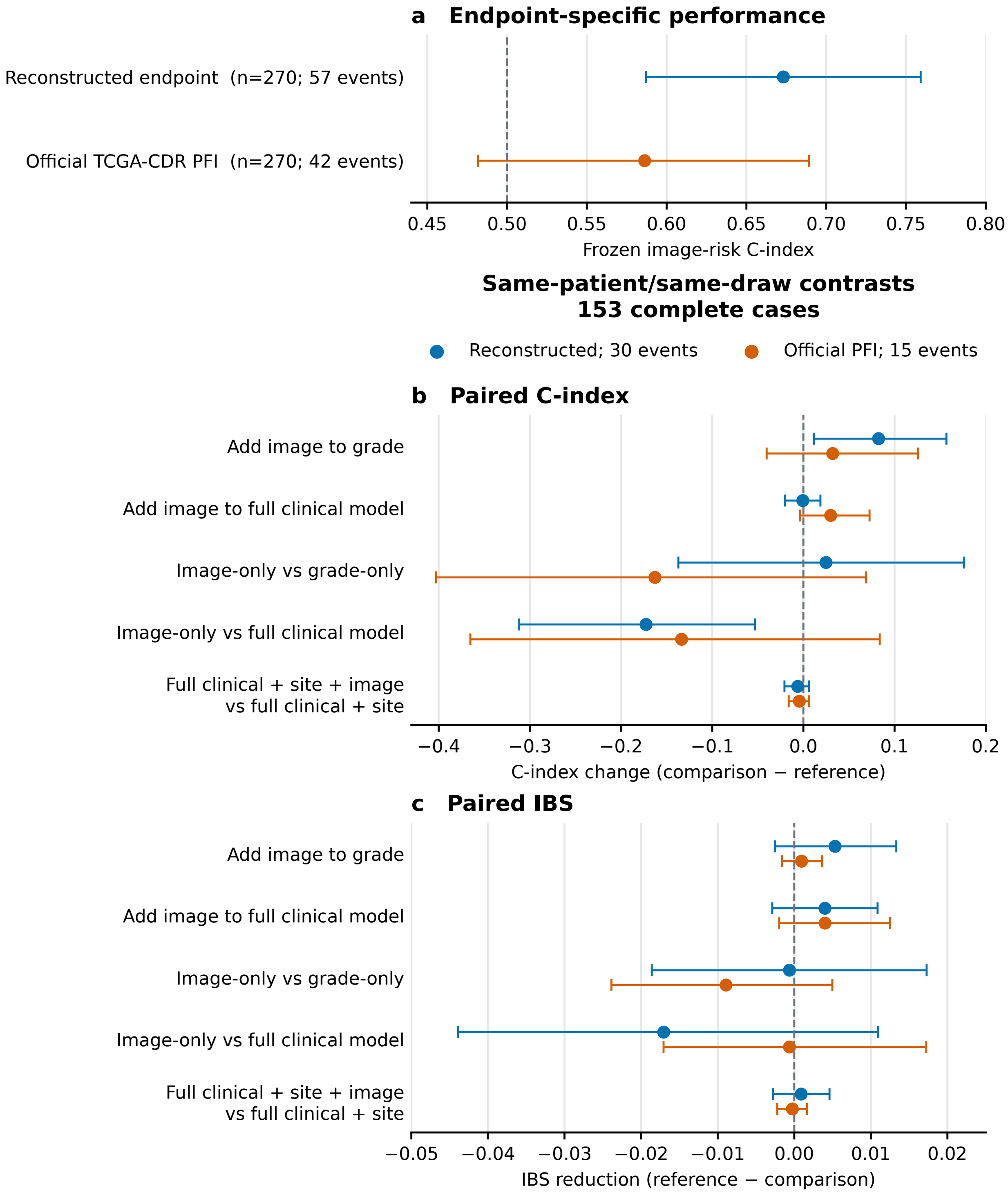

Figure 5: Endpoint-conditioned marker 7 transfer. **a**, Frozen image-risk C-index in 270 TCGA-PRAD patients for the reconstructed with-tumor endpoint (57 events) and Official TCGA-CDR progression-free interval (PFI; 42 events). **b**–**c**, Same-patient, same-bootstrap-draw paired changes in C-index and integrated Brier score (IBS; 0.5–5 years) for five model contrasts among 153 complete cases (30 reconstructed-endpoint and 15 official-PFI events). Blue denotes the reconstructed endpoint and orange official PFI. Points and whiskers are estimates and 95% percentile patient-bootstrap intervals from 2,000 valid paired draws; zero draws were undefined. Dashed lines mark 0.5 in **a** and no change in **b**–**c**. C-index change is comparison minus reference and IBS improvement is reference minus comparison, so positive values favor the first-named comparison model. The endpoint definitions are not equivalent.

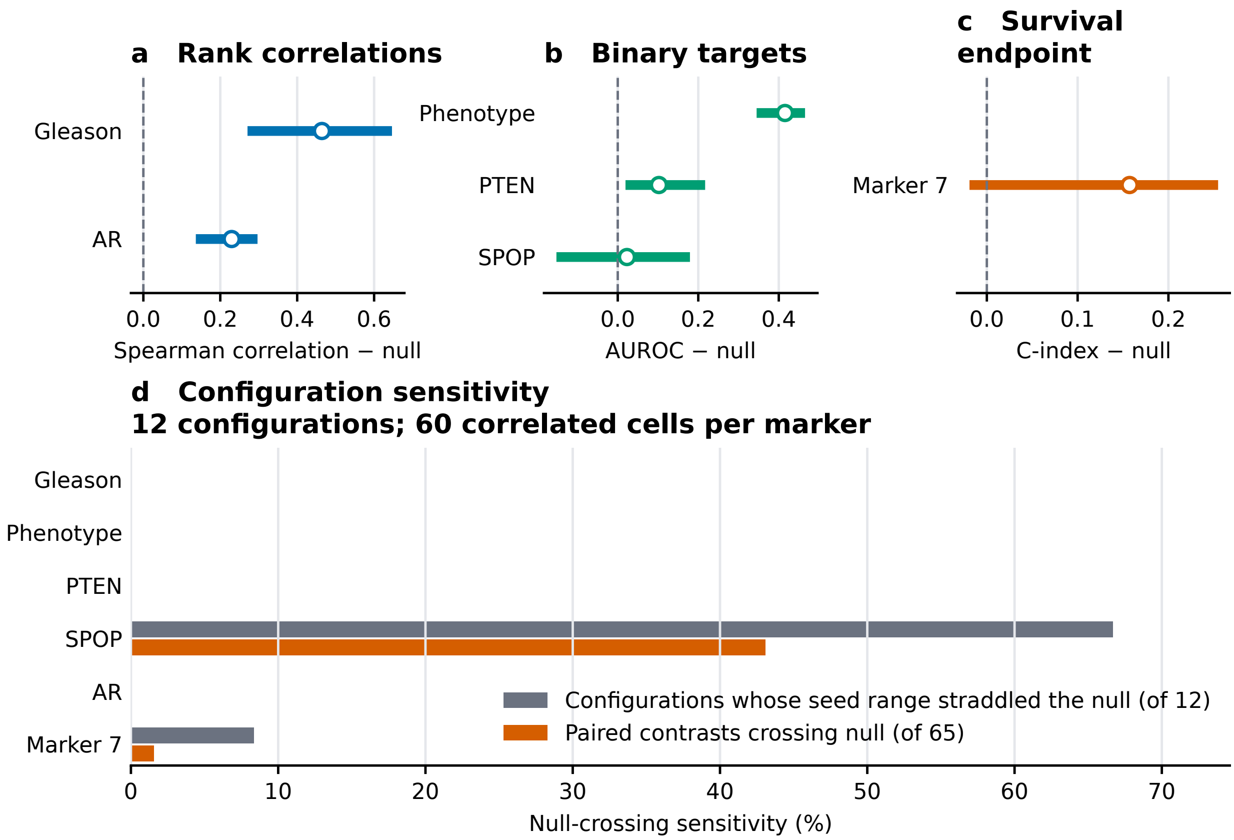

Figure 6: Decision-level stability overview. **a–c**, Spearman correlation, AUROC, and C-index summaries expressed relative to their marker-specific null. Thick lines span the global minimum–maximum across 60 correlated seed cells per marker (12 encoder–scale–tile configurations, five seeds each); open circles mark the 60-cell mean and dashed lines the null. **d**, Percentage of the 12 configurations whose observed five-seed range straddled the null (grey) and percentage of 65 seed-matched paired contrasts whose endpoints lay on opposite sides of the null (orange). The cells reuse patients and folds, so neither the ranges nor the null-crossing percentages are population uncertainty estimates or independent validations. Complete grids and contrast directions are provided in the Supplementary Information.

signal changed with endpoint, covariate hierarchy and representation. The correlated grid further showed that repetition across settings could either reinforce a direction or expose conditionality, but could not replace external data.

The resulting evidence map (Table 1) is the primary synthesis of the study. It records the strongest support justified for each target together with the axis that remains unresolved. Its purpose is not to rank encoders or biomarkers, but to prevent a pooled association from being promoted beyond the evidence that qualifies it.

Qualification decision. The third proposition was supported: target-level evidence states conveyed distinctions that a single performance ranking would conceal. These states remain research qualifications, not claims of clinical validity or biological absence.

Table 1: Joint evidence map after target-specific qualification audits.

Evidence question	Evidence state	Supported interpretation
Qualification logic	descriptive framework	marker-specific
Grade and phenotype transport	transportable	Externally transportable
PTEN qualification	context sensitive	Internally supported; multisite-stable
SPOP qualification	unsupported in frozen design	Primary unsupported; configuration-sensitive
AR qualification	context sensitive	Pooled direction supported; site transportability unresolved
Recurrence transfer	context sensitive	Post-hoc exploratory; endpoint/encoder/scale-limited
Recurrence increment	context sensitive	Exploratory; endpoint-conditioned
Setting sensitivity	descriptive framework	Descriptive sensitivity evidence

3 Discussion

3.1 The contribution is a qualification logic, not another performance benchmark

The central contribution of this study is to separate signal detection from signal qualification. Frozen pathology representations made every evaluated association technically testable, but the scientific status of those associations differed once transport, confounding, site, representation and endpoint were examined separately. The result is not a league table of encoders or targets. It is an operational logic for deciding which claim the available evidence supports, which claim must be restricted to a context, and which claim remains unsupported in the tested design. The accompanying evidence map makes that decision explicit and source-linked rather than leaving qualification implicit in a collection of performance estimates.

3.2 The three propositions are supported at different evidentiary levels

The first proposition was supported within the available cross-cohort comparisons: grade and phenotype, which are directly expressed in routine morphology, showed the clearest transport. This finding does not establish that morphologic targets are universally easier than molecular targets, because matching external labels were not equally available. Rather, it identifies the strongest evidence state attained in this study and demonstrates what transport evidence looks like when a frozen probe is applied without refitting.

The second proposition was supported by the narrowing of apparently positive signals. PTEN retained within-cohort support but not an established held-out increment beyond grade. AR retained a positive pooled direction while site transport and grade-independent increment remained unresolved. The post-hoc recurrence signal changed with endpoint, covariate hierarchy and representation. SPOP illustrates a different boundary: the fixed primary analysis was unsupported and the wider audit was setting-sensitive, but neither result proves biological absence. Together, these cases show that target morphology, clinical structure and data provenance determine the scope of a foundation-model claim.

The third proposition was supported by the separation of targets into transportable, context-sensitive and unsupported evidence states. These states retained information that a scalar ranking would discard: the same pooled direction could coexist with uncertain increment, unresolved site transport or endpoint-dependent transfer. The framework is thus a decision structure for research evidence, not a new significance test or a calibrated probability that a biomarker is valid.

3.3 Why pooled performance fails as a scientific claim

Pooled performance fails when it answers a narrower question than the claim attached to it. A pooled association can show that information is present in a representation without showing what generated that information. Grade composition may explain part of a molecular association; institutional composition may support a pooled effect that is imprecise within sites; encoder and scale choices may change the recovered signal; and similarly named clinical outcomes may encode different events. None of these failures makes the original metric erroneous. They make the broader interpretation underidentified.

This distinction changes how robustness should be reported. Repetition across encoders, scales, tile budgets and seeds can reveal dependence on analytic choices, but the settings reuse patients and folds and therefore cannot manufacture independent replication. Likewise, reproducible code and source-linked outputs establish computational traceability, not external validity. The practical implication is that a candidate signal should advance only with the evidence state and unresolved axes reported together. Performance without that qualification invites the strongest-looking number to stand in for transport, increment and clinical meaning that were never tested.

3.4 Limitations and unresolved inferential problems

The main limitation is asymmetry of evidence. Grade and phenotype had compatible cross-cohort labels, whereas molecular targets were available primarily within TCGA-PRAD. The stronger state attained by morphology-linked targets therefore reflects both their observed transport and the opportunity to test transport. It cannot establish an inherent ordering of biological learnability across target classes. The framework also evaluates only the axes represented in the available data; unmeasured clinical, demographic, pre-analytic or acquisition factors may remain.

Several uncertainty estimates were inherited from legacy outputs. Some saved intervals did not retain valid and undefined bootstrap-replicate accounting, and some binary rows did not retain event counts. Those quantities were not reconstructed. The configuration grid reused cohorts, folds and patients, so its seed intervals describe tile-sampling variation rather than patient or population uncertainty. Its cells and paired contrasts are correlated and cannot be counted as independent confirmations. Consequently, stability across the grid can bound setting dependence but cannot substitute for a new cohort.

The site analysis cannot separate tissue source from scanner, staining, preparation or case mix. Site-specific uncertainty therefore remains a transport problem rather than evidence for a causal acquisition mechanism. Recurrence analyses face a related identification problem: reconstructed recurrence, progression-free measures and official PFI use different event definitions and are not interchangeable. The paired analyses used a complete-case cohort with modest event counts and no multiple imputation, so selection and missingness may affect the estimated increments.

The recurrence signal was discovered post hoc, and its PCA sweep was not nested model selection. It therefore remains exploratory even where a point estimate appears favorable. More generally, the three evidence states are qualitative research decisions tied to the frozen analyses and available labels; they are not posterior probabilities, universal biomarker grades or clinical action thresholds. The study covers prostate pathology, two frozen encoders and the represented institutions. It does not establish whole-slide diagnostic performance, treatment utility, population prevalence, causal biology or generalization to other cancers and deployment environments.

3.5 Implications and conclusion

The unresolved axes specify the next experiments more clearly than a request for more benchmarking. A transport claim requires an independent cohort with a compatible target definition and locked preprocessing. A molecular claim requires target-appropriate assays and patient-level evaluation beyond grade and other clinical structure. A site claim requires acquisition metadata or a design that can separate site from scanner, stain and case mix. An outcome claim requires harmonized endpoints, adequate events, model selection locked before target-cohort evaluation and paired incremental comparison against a clinically meaningful reference. These are not optional refinements to a pooled metric; they are the evidence needed to change its qualification state.

The study therefore establishes a concrete rule for pathology foundation-model research: do not promote a detectable association on pooled performance alone. Report whether it transports, what clinical information it adds, how it behaves across sites and representations, which endpoint it predicts, and which dimension remains unresolved. In the evaluated data, that logic supported grade and phenotype transport, narrowed PTEN and AR, left SPOP unsupported in the fixed design, and restricted recurrence to an exploratory, endpoint- and representation-conditioned interpretation. The broader contribution is an auditable framework and empirical failure map for prioritizing candidate signals without claiming more than the evidence can support.

4 Methods

The Methods follow the same evidentiary sequence as the Results. We first define cohorts, labels, analysis units and frozen representations; then specify cross-cohort and patient-disjoint evaluation; and finally describe confounder, site, setting and endpoint audits, uncertainty and provenance. This ordering keeps each qualification decision tied to the data and analysis that can support it.

4.1 Cohorts and labels

NADT-Prostate provided paired prostate histology, per-core or per-slide grade and phenotype labels, and patient identifiers for source-probe fitting and patient aggregation. PANDA provided institution-labelled case images and official ISUP grades; its tumor/benign phenotype was derived as ISUP 0 versus ISUP at least 1. TCGA-PRAD provided histology, patient identifiers, grade, PTEN copy-number loss, SPOP mutation and AR-activity labels, as well as separately constructed recurrence-style endpoints. PRECISE provided expert-annotated serial sections and per-image pathologist Gleason scores. LEOPARD provided biochemical recurrence event and follow-up labels for recurrence-model fitting. The public NADT, PANDA and PRECISE resources are described in their source records [4, 3, 10]; LEOPARD is described in the MIDL 2026 challenge report [8], and the organizer baseline remains an arXiv preprint [9]. Each analysis used only rows with an evaluable target; missing or uncertain outcomes were not converted to negative labels.

For NADT patient-level evaluation, labels and predictions were aggregated within patient; the phenotype target was the patient's mean tumor fraction rather than a rounded binary label. PANDA sampling capped each institution–ISUP stratum at 100 cases before tissue filtering; its grade-derived phenotype was not a separately supplied annotation.

The recurrence endpoints were retained under their source definitions. The Reconstructed with-tumor endpoint used a recorded with-tumor follow-up as an event and censored at the latest valid non-event follow-up. The strict recurrence-only sensitivity required a tumor-free visit before a later with-tumor visit. TCGA-CDR PFS and DFS used their named cBioPortal comparator fields. Official TCGA-CDR PFI used the official PFI event and time fields [11]. No endpoint substitution was renamed as official PFI.

4.2 Frozen embeddings and aggregation

CONCH and Virchow were used as frozen encoders; encoder weights were not fine-tuned. Tissue-qualified tiles were embedded and averaged to slide representations. Where the endpoint unit was the patient, multiple slide values were aggregated by patient before primary scoring. PANDA used case-image rows, PRECISE used imaging sessions, AR site effects used slide-level estimates with patient-cluster resampling, and all such departures from a patient-level effect unit were recorded in the figure source tables.

The sensitivity analysis crossed each encoder with native and shared physical scales, tile budgets of 16, 32 and 64, and five repeated sampling seeds. These settings were applied to the fixed cohorts and folds. They were not treated as newly sampled populations.

4.3 Fixed probes and patient-disjoint evaluation

Continuous marker-family targets used standardized ridge regression with the saved fixed alpha grid. Binary TCGA-PRAD screening tests used standardized logistic regression with fixed regularization and balanced class weighting. The frozen NADT phenotype probe used for PANDA transfer was instead `LogisticRegression(C=1.0)` with 2,000 maximum iterations and without class weighting, matching its saved generating script. Within-cohort evaluation used five patient-disjoint grouped folds, keeping all slides from a patient within one fold. Continuous targets were scored by patient Spearman correlation, binary targets by patient AUROC, and time-to-event targets by patient-level concordance and the saved time-dependent metrics.

For zero-shot transfer, the source-cohort probe and preprocessing were applied to the target cohort without refitting. Native refits and held-out site analyses were reported as distinct evaluation types and did not substitute for zero-shot transfer.

4.4 Qualification criteria and multiplicity

Transportable evidence required an appropriate cross-cohort evaluation with consistent direction. Context-sensitive evidence indicated material dependence on site, encoder, physical scale, tile budget or endpoint. Unsupported evidence indicated that the frozen primary analysis did not support the proposed effect; it was not interpreted as evidence of biological absence. These dimensions address published concerns about molecular-label confounding, task-dependent foundation-model rankings and institution-associated technical signals [5, 6, 7].

The saved 17-test family comprised 13 marker hypotheses—the six initial candidates and seven additional screening candidates—plus four nested/refit confounder audits: PTEN beyond grade, AR beyond grade, and marker 7 beyond grade or a fully adjusted clinical model. Continuous marker hypotheses used two-sided Spearman tests; binary marker hypotheses used two-sided Mann–Whitney tests. The fully refitted, grade-stratified audit permutations used one-sided refit-permutation p values for positive improvement. Statistical support used $\alpha = 0.05$ with Benjamini–Hochberg false-discovery-rate adjustment across this family. Re-tests of an existing hypothesis in another cohort or encoder were qualification evidence and were not added to the discovery family.

4.5 Confounder and site analyses

PTEN and AR increment analyses compared clinical-only, image-only and combined models in outer patient-disjoint folds. Inner out-of-fold image predictions were used to construct outer-training meta-model features, and the image probe was refitted on each complete outer-training fold before application to untouched outer-test patients. Primary increments were the paired held-out change in AUROC for PTEN and change in R^2 for AR.

The nonparametric check permuted patient labels within rounded patient-mean Gleason strata and refitted the full nested pipeline. AR site analyses estimated slide-level Spearman correlations and resampled patients as clusters. Site descriptors and pooled estimates were kept at their recorded units. Raw scanner identifiers were not interpreted as validated scanner models, and absent stain metadata were not inferred.

4.6 Correlated stability analysis

For each marker, seed-level scores were summarized into encoder–scale–tile configurations. The saved summaries report mean, sample standard deviation, range, chance-or-worse count and a Student-t interval across the five tile-sampling seeds. AUROC and C-index used a null of 0.5; Spearman correlation used zero. Paired comparisons preserved the seed-level pairing for scale, encoder and tile-budget contrasts.

Because cohorts, patients and folds were shared across settings, all grid summaries were interpreted as correlated sensitivity analyses. Seed intervals were descriptive sampling intervals, not patient-bootstrap or population-confidence intervals. No grid setting was used to revise frozen labels, scores, folds or evidence thresholds.

4.7 Recurrence endpoints and paired comparisons

Marker 7 was a post-hoc Cox risk score fitted in LEOPARD and applied without target-cohort refitting to TCGA-PRAD. Its existing PCA dimension sweep served only as exploratory sensitivity analysis. PCA-dimension selection was not nested within an outer evaluation, so the sweep did not provide an unbiased model-selection estimate and no dimension was interpreted as an unbiased optimum.

Paired TCGA-PRAD comparisons used the same complete-case patients and the same bootstrap draw for the clinical model and its image-augmented counterpart. Models were compared for the Reconstructed with-tumor endpoint and Official TCGA-CDR PFI separately. C-index improvement was defined as augmented minus reference C-index; integrated-Brier-score improvement was reference minus augmented IBS, so positive values favored the augmented model for both metrics. Reconstructed recurrence, PFS, DFS and official PFI were never pooled into one endpoint.

4.8 Bootstrap uncertainty and missingness

Where the saved source supported this construction, patient-level estimates used 95% percentile patient-bootstrap intervals. AR site rows used a patient-cluster bootstrap around a slide-level effect. For legacy NADT/TCGA primary and AR site interval summaries, saved tables retained interval endpoints but not requested, valid and undefined accounting, except requested 2,000 where explicitly recorded; unavailable counts were not reconstructed. Paired marker 7 outputs separately retained 2,000 valid and zero undefined replicates. Paired survival contrasts used identical patient draws for both models. Outputs with replicate accounting retained and reported undefined replicate counts and fractions; estimates without a saved interval were marked unavailable. Analyses used their saved evaluable or complete-case frames; no missing, uncertain or unevaluable value was recoded as a negative outcome. Multiple imputation was not performed for the marker 7 paired comparison.

4.9 Software and provenance

Analyses were executed with the repository's locked Python environment and auditable entry scripts. Fixed random seeds, software versions, input hashes, endpoint mappings and output hashes were recorded in saved run configurations and manifests. Main figures were rendered from saved CSV source tables rather than transient objects. Source-table schema, analysis unit, denominator, endpoint identifier and interval type were checked before a result was used in the manuscript. The clinician review source for PRECISE remained immutable throughout the manuscript work.

Large language model assistance. An assistive large language model, OpenAI Codex, was used during code and manuscript revision for code editing, test authoring, figure-production assistance and manuscript language revision. It was not treated as an author and did not independently determine the study endpoints or scientific conclusions. The exact product/model version, usage dates and accountable authors' final verification statement require author confirmation before submission.

Draft status—author confirmation required. The availability statements below describe current sources and local artifacts. Final author contributions, competing interests, funding, institutional ethics and consent wording, release accessions and submission metadata remain author-controlled blocking items; this draft does not represent them as finalized declarations.

Data Availability

The public cohorts used in this study retain their source-specific access and reuse terms. NADT-Prostate is available from The Cancer Imaging Archive (DOI: 10.7937/TCIA.JHQD-FR46) [4]; PANDA data are distributed through the PANDA challenge under its platform terms [3]; TCGA-PRAD histology and molecular data are available through the Genomic Data Commons, with the referenced TCGA clinical endpoint fields available through cBioPortal; and PRECISE is available from Zenodo (DOI: 10.5281/zenodo.20721779) [10]. LEOPARD is an access-governed challenge resource: its training data and outcome labels remain subject to the LEOPARD challenge and dataset terms, and its public challenge report is cited here [8]. Publication-facing aggregate numerical figure-source tables are supplied with the public code repository. Patient-level derived analysis artifacts and local provenance manifests are not redistributed. No separate derived-data archive DOI has yet been assigned; the final deposition and accession statement must be confirmed before submission.

Code Availability

Source code for the audited analyses and manuscript generation is publicly available in the manuscript development branch at `https://github.com/AiXLab-GNU/vlm-pathology/tree/feat/precise-pni-morphology-rereview`. Principal auditable analysis entry points are

- `models/build_revision_p0_artifacts.py`;
- `models/aggregate_stability_grid.py`;

- `models/build_tcga_cdr_pfi_evidence.py`;
- `models/build_marker7_survival_paired_analysis.py`;
- `models/run_marker7_common_source_sensitivity.py`; and
- `models/build_ar_spop_evidence_closure.py`.

The repository also versions the stability runners, manuscript builder, figure renderers, table generators, validation tests and environment specifications. Manuscript figures are regenerated from saved CSV source tables by the scripts in `paper/figures/`. Repository-authored software and documentation are released under the MIT License. Dataset copies, whole-slide images, cached embeddings, pretrained model weights, access-governed LEOPARD artifacts and patient-level derived outputs are not redistributed; they must be obtained or reconstructed from their original sources under the applicable access terms and are not covered by the repository's software license. No versioned archival DOI was available at the time of manuscript preparation.

Author Contributions

No contribution roles were inferred from repository contents. A contribution statement using the final author list and author-approved roles must replace this factual draft note before submission.

Competing Interests

No competing-interest status was inferred. Each named author's explicit declaration of interests, including an explicit no-interest statement where applicable, must be confirmed before submission.

Ethics and consent statement

This work was a secondary analysis of deidentified data and involved no new participant recruitment, intervention or specimen collection. The original participant approvals, consent procedures and data-use conditions remain those reported by the source publications and repositories. The corresponding author must confirm the applicable institutional determination for this secondary analysis and the final approval, waiver and consent wording before submission; no institutional decision was inferred in this draft.

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